1D CNN — Architecture Variant Comparison

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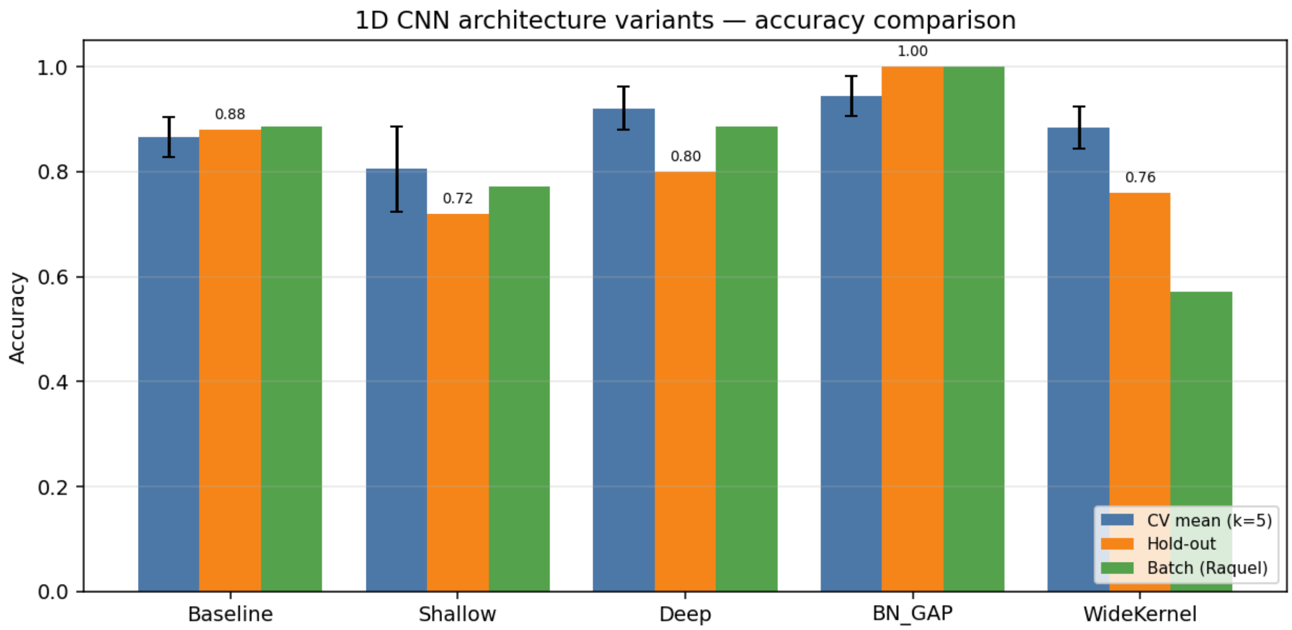
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Total trials	241 (train pool 216, hold-out 25)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

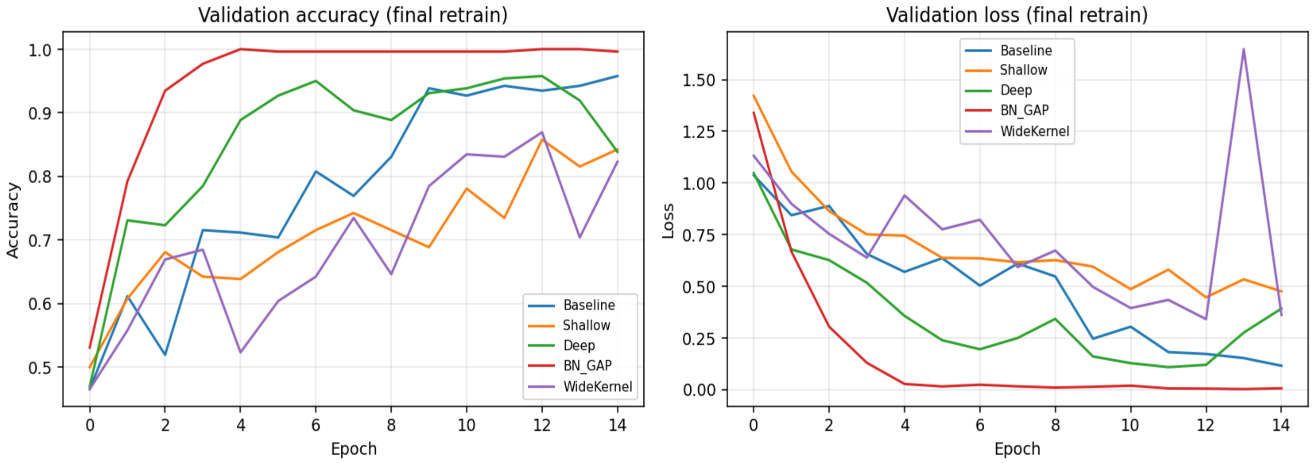
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.8656	0.0375	0.8800	0.3752	0.8857	31/35
Shallow	72,487	0.8056	0.0812	0.7200	0.8611	0.7714	27/35
Deep	204,007	0.9211	0.0407	0.8000	0.4292	0.8857	31/35
BN_GAP	57,319	0.9444	0.0378	1.0000	0.0239	1.0000	35/35
WideKernel	131,687	0.8839	0.0393	0.7600	0.4297	0.5714	20/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Validation accuracy (final retrain)
Accuracy
Epoch
Baseline
Shallow
Deep
BN_GAP
WideKernel
Validation loss (final retrain)
Loss
Epoch
Baseline
Shallow
Deep
BN_GAP
WideKernel

Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.8656 ± 0.0375
Hold-out test acc / loss	0.8800 · 0.3752
Batch-test acc	0.8857 (31/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.8864	0.9545	0.8864
2	1038	43	0.8837	0.9535	0.8837
3	1038	43	0.8837	0.8837	0.8837
4	1038	43	0.7907	0.8837	0.7907
5	1038	43	0.8837	0.8837	0.8837

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.800	1.000	0.889	4
Double_Nothing	0.500	0.667	0.571	3
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	0.333	0.500	3
macro avg	*0.900*	*0.857*	*0.851*	*25*
weighted avg	*0.908*	*0.880*	*0.871*	*25*

Batch test per class (Raquel)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	3	5	0.600
Overall	**31**	**35**	**0.886**

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8056 ± 0.0812
Hold-out test acc / loss	0.7200 · 0.8611
Batch-test acc	0.7714 (27/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.7955	0.8864	0.7955
2	1038	43	0.9302	0.9302	0.9302
3	1038	43	0.7209	0.7442	0.7209
4	1038	43	0.7209	0.7209	0.7209
5	1038	43	0.8605	0.8605	0.8605

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.750	0.857	4
Double_Nothing	0.200	0.333	0.250	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.333	0.333	0.333	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	0.333	0.500	3
macro avg	*0.762*	*0.679*	*0.690*	*25*
weighted avg	*0.792*	*0.720*	*0.729*	*25*

Batch test per class (Raquel)

Class	Correct	Total	Accuracy
Double_Lower	2	5	0.400
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	2	5	0.400
Overall	**27**	**35**	**0.771**

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9211 ± 0.0407
Hold-out test acc / loss	0.8000 · 0.4292
Batch-test acc	0.8857 (31/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9545	0.9545
2	1038	43	0.9767	0.9767	0.9767
3	1038	43	0.9070	0.9535	0.9070
4	1038	43	0.8605	0.8837	0.8605
5	1038	43	0.9070	0.9070	0.9070

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.667	1.000	0.800	4
Double_Nothing	0.000	0.000	0.000	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.600	1.000	0.750	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	0.333	0.500	3
macro avg	*0.724*	*0.762*	*0.706*	*25*
weighted avg	*0.747*	*0.800*	*0.740*	*25*

Batch test per class (Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	3	5	0.600
Overall	**31**	**35**	**0.886**

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9444 ± 0.0378
Hold-out test acc / loss	1.0000 · 0.0239
Batch-test acc	1.0000 (35/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9773	0.9545
2	1038	43	0.9535	1.0000	0.9535
3	1038	43	1.0000	1.0000	1.0000
4	1038	43	0.9302	0.9302	0.9302
5	1038	43	0.8837	0.9535	0.8837

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	1.000	1.000	3
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	*1.000*	*1.000*	*1.000*	*25*
weighted avg	*1.000*	*1.000*	*1.000*	*25*

Batch test per class (Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	**35**	**35**	**1.000**

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8839 ± 0.0393
Hold-out test acc / loss	0.7600 · 0.4297
Batch-test acc	0.5714 (20/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9545	0.9545
2	1038	43	0.8837	0.9302	0.8837
3	1038	43	0.8837	0.9070	0.8837
4	1038	43	0.8372	0.8837	0.8372
5	1038	43	0.8605	0.8837	0.8605

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.250	0.400	4
Double_Nothing	0.000	0.000	0.000	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.429	1.000	0.600	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	0.750	1.000	0.857	3
macro avg	*0.711*	*0.750*	*0.678*	*25*
weighted avg	*0.749*	*0.760*	*0.701*	*25*

Batch test per class (Raquel)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	4	5	0.800
Drum_Roll	0	5	0.000
Overall	**20**	**35**	**0.571**